

Setup

I generated 100 unconditional protein backbone structures (50 from La Proteina and 50 from ReQFlow), each ~200 residues long. Structures were evaluated using CCTBX-based Ramachandran analysis, with optional clashscore computation using reduce and probe.

Challenges and Solution

- **Environment setup (CCTBX + reduce/probe):** Installing and configuring validation tools across systems was inconsistent. This was resolved by using a conda environment and documenting fallback steps for systems where reduce/probe were unavailable.
- **Clashscore pipeline complexity:** Clashscore requires hydrogen preprocessing, which introduces additional steps and failure points. This was handled by making clashscore optional and ensuring the pipeline runs reliably with Ramachandran metrics alone.
- **Handling malformed or inconsistent PDB files:** Some generated structures caused parsing issues. Robust error handling was added to ensure validation continues even if individual files fail.

Summary Statistics

Method	Mean Favored	Median Favored	Mean Outlier	Median Outlier	Successful
Laproteina	0.9944	0.9949	0.000200	0.000000	50/50
Reqflow	0.9781	0.9798	0.004600	0.005100	50/50

Here, Mean Favored = Mean Rama Favored Fraction, same goes for median
All 100 structures validated successfully, with no steric clashes detected.

Key Insights

- **La Proteina produces higher-quality backbones:** It achieves a higher favored fraction and lower outlier fraction, indicating better adherence to preferred backbone conformations.
- **ReQFlow shows greater variability:** although it still produces valid structures, its wider distribution suggests less consistent backbone geometry.
- **Both methods generate physically plausible structures:** Favored fractions are close to or above the ~98% threshold typical of high-quality protein structures.
- **No steric clashes observed:** Both models produce geometrically valid structures without atomic overlaps, indicating good local physical consistency.

Conclusion

Both La Proteina and ReQFlow successfully generate physically realistic protein backbones. However, La Proteina demonstrates more consistent structural quality, while ReQFlow offers greater variability. Overall, the validation confirms that both methods produce biologically plausible backbone conformations suitable for downstream analysis.